Mikolaj Hilgert

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Skills

Development & Data Tools:	Python, C# (.NET), Java (Spring), TypeScript (NestJS), Scala, Spark, Airflow, LangChain, Pinecone, Postgres, MongoDB, NextJS
Cloud & Infrastructure:	Microsoft Azure, Google Cloud, Docker, Kubernetes, Git, Terraform, DevOps, MLOps
Languages	English (native), Polish (native), Czech (proficient), German (B1)

Experience

Graduate Data Science Intern (Full-Time) — **Nov 2025 — Present**
ASML — *Veldhoven, The Netherlands*

- Master's thesis internship project on the topic of predictive maintenance of high-tech lithography equipment.

Software Developer (Part-Time) — **Sep 2024 — Dec 2025**
CY2 — *s-Hertogenbosch, The Netherlands*

- Managed the development of a Microsoft Teams–based student app that unifies LMS, SIS, and other systems, giving students and staff a single interface to access and act on information.
- Developed a multi-tenant, LLM-powered chatbot platform that empowers educational organizations to create tailored chatbots using their own documents and datastores.

Data Science Intern (Full-Time) — **Feb 2024 — Jul 2024**
ASML — *Veldhoven, The Netherlands*

- Thesis title: Impact of parallel unscheduled machine downs on the eXtreme long down (XLD) frequency of DUV lithography systems.
- Conducted data analysis and developed an algorithm to quantify the impact of staffing on XLD frequency.
- Researched and implemented machine learning techniques to predict XLD occurrences.

Trainee Software Engineer (Part-Time) — **Mar 2023 — Nov 2023**
CM.com (prev. Building Blocks) — *Tilburg, The Netherlands*

- Worked on improving the AI Decisioning Engine, which is used to deliver mass consumer data personalisations, increase conversion rates, order value, and lower returns for clients like Coca-Cola, Corendon, Blokker, and Basic-Fit.
- Enhanced the IntelliJ IDEA development plugin, which was utilised on a daily basis by over 20 data scientists, expediting the development cycle and streamlining the delivery of tailored solutions for all Building Blocks' clients.

Software Engineering Intern (Full-Time) — **Sep 2022 — Feb 2023**
CM.com (prev. Building Blocks) — *Tilburg, The Netherlands*

- Implemented an advanced user-detail encryption module utilising Apache Spark within an existing production codebase, improving data security and GDPR compliance.
- Developed a data loader for Building Blocks' proprietary platform for files stored in Azure Blob Storage.

Research Student Intern (Part-Time) — **Feb 2019 — Jun 2019**
Microsoft — *Prague, Czechia*

- Conducted Market research on Quantum computing and presented findings for an internal Hackathon.

Notable projects

Cloud-Native Full-Stack Application — **Sep 2023 — Jan 2024**
Individual University project

- An online platform that connects data providers with skilled labellers, streamlining image data labeling to create high-quality data-sets that are used in AI and machine learning applications.
- Developed a microservice architecture in C# using .NET 6.0, with an automated CI/CD pipeline including testing and deployment to a managed Kubernetes cluster in Microsoft Azure.

Full-Stack Dashboard with ETL Data Pipeline Processing — **Sep 2021 — Jan 2022**
Group University project

- Collaborated in a high-performing team of 5 students following the Scrum Agile framework.
- Contributed to the design and development of the ETL data pipeline using Python and Apache Airflow.

Education

Master of Science in Data Science in Business and Entrepreneurship, *JADS (Tu/e & Tilburg University)* — Aug 2024 — Jul 2026
Pre-master in Data Science in Business and Entrepreneurship, *JADS (Tu/e & Tilburg University)* **Grade: 8.0** — Jan 2023 — Jun 2023
Bachelor of Science in ICT & Software Engineering, *Fontys University of Applied Science* **Grade: 9.2** — Sep 2020 — Jul 2024